Deep Learning-Driven Crack Quantification and Remaining Useful Life Prediction for Dam Structural Health Monitoring

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Abstract — Structural health monitoring (SHM) of concrete dams needs reliable and automatic methods for measuring crack sizes. This helps evaluate structural soundness. Inspecting cracks by hand with rulers and gauges takes a lot of time and is not practical. This paper introduces a complete deep learning system that uses a pretrained ResNet50 convolutional neural network (CNN), along with Linear Elastic Fracture Mechanics (LEFM) and Paris' Law, to automatically measure crack width and predict RUL from field images.

The model is trained on the Virginia Tech Crack Quantification ROIs Dataset, which includes 45 actual crack images taken at three distances. We augment this dataset to 450 samples using nine types of photometric and geometric changes. The CNN achieves a mean absolute error of 0.48 mm for average crack width and 1.27 mm for maximum crack width on the separate test set. The predicted crack sizes are combined with a fracture mechanics engine to calculate the Mode-I stress intensity factor K_I and estimate RUL through numerical time integration. This system is non-invasive, scalable, and works well with drone or camera-based inspection methods for ongoing dam health evaluations.

Keywords: structural health monitoring, crack width estimation, convolutional neural network, remaining useful life, fracture mechanics, Paris's Law

1. Introduction

Concrete gravity dams form a critical component of national water and energy infrastructure. Their structural integrity is continuously challenged by hydrostatic loading, seismic activity, freeze-thaw cycling, alkali-silica reaction, and material fatigue. Surface cracking is one of the earliest and most visible forms of structural degradation. Left undetected or underestimated, cracks can propagate under cyclic hydraulic pressure to cause catastrophic failure with significant loss of life and property.

Current inspection practice relies on periodic manual surveys in which engineers use crack rulers, optical comparators, and feeler gauges to measure crack width at discrete locations. Autonomous inspection platforms , unmanned aerial vehicles equipped with high-resolution cameras are increasingly deployed for dam surveillance, generating large volumes of crack imagery that cannot be manually processed at the required frequency.

Deep learning has shown strong performance in automated crack detection and segmentation across infrastructure types including pavements, bridges, and buildings. However, most existing work addresses binary crack/no-crack classification or pixel-level segmentation rather than the quantitative dimensional measurement required for fracture mechanics analysis. The link between deep learning-based crack measurement and physics-based RUL prediction remains largely unexplored.

This paper addresses this gap by proposing a complete pipeline that trains a ResNet50-based regression CNN to predict mean and maximum crack width from raw field images, derives crack length and orientation from binary segmentation masks using classical computer vision, integrates all predicted dimensions with LEFM to compute the stress intensity factor, and applies Paris' Law numerical integration to estimate RUL. The pipeline is validated on the Virginia Tech Crack Quantification ROIs Dataset [1], a publicly available collection of field crack images with calibrated width measurements.

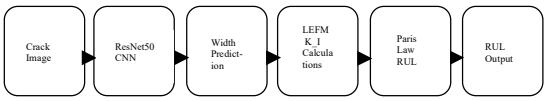

Fig. 1: End-to-end pipeline from crack image to RUL estimate.

2. Literature Review

Early deep learning approaches to detecting pavement cracks used simple CNN classifiers that operated on fixed-size image patches [2]. Later, encoder-decoder architectures like U-Net and DeepLab allowed for pixel-level crack segmentation with high recall. DeepCrack [3] introduced hierarchical feature fusion for identifying thin cracks. However, these methods produce binary masks instead of actual measurements.

Crack width estimation from images has not been studied as much. Zou et al. [4] applied skeleton analysis on segmentation masks to estimate widths. Cha et al. [5] used Faster R-CNN for crack detection but did not predict size. More recent studies have utilized keypoint detection and regression to predict crack width at certain locations, but they have not integrated this with structural mechanics models.

In fracture mechanics, Paris and Erdogan [6] established the empirical power-law relationship between the stress intensity factor range and crack growth rate. This relationship is the basis for predicting fatigue life. Researchers have extensively studied LEFM-based RUL models for concrete in dam engineering literature [7], but these models usually assume that crack sizes are known from manual inspections instead of vision-based measurements. This work connects these two research areas.

3. Methodology

3.1 Dataset

The Virginia Tech Crack Quantification ROIs Dataset [1] contains 45 field images of 15 distinct concrete crack specimens photographed at standoff distances of 0.5 m, 0.75 m, and 1.0 m using a high-resolution camera with circle calibration markers. For each image, five Regions of Interest (ROIs) were manually selected and crack widths were measured using a calibrated ruler, yielding ground truth values ranging from 0.15 mm to 8.5 mm. Binary segmentation masks annotated by domain experts are provided for each image and standoff distance combination.

3.2 Data Augmentation

With only 45 images, overfitting is a primary concern. Nine augmentation strategies were systematically applied to each training image, expanding the dataset to 450 samples. Table 2 summarizes the augmentations applied. Width labels are invariant to all photometric transforms; orientation labels are corrected for geometric transforms (flips and rotations) following standard angle adjustment rules.

Table 2: Augmentation strategies applied to training images.

Augmentation Type	Description
H/V Flip	Horizontal & vertical mirror
Rotation	90°, 180°, 270°
Brightness	0.6× and 1.4× scaling
Contrast	0.6× and 1.5× scaling
Gamma	0.7 and 1.3
Gaussian Blur	kernel 5×5, σ∈[0.1,2.0]
Sharpness	2.0× enhancement
Random Erasing	p=0.2, scale 2–10%
Colour Jitter	brightness, contrast, hue
Total: 9 augmentations × 45 = 450 images	

3.3 CNN Architecture

A ResNet50 backbone pretrained on ImageNet-1K serves as the feature extractor. The original 1000-class fully connected head is replaced with a shared 512-dimensional layer (LayerNorm, GELU activation, dropout p=0.5) fed by the 2048-dimensional global average pool output. Two independent regression heads branch from the shared layer, one for mean width and one for maximum width prediction. Each head applies:

$$\hat{y} = Softplus(W_2 \cdot GELU(W_1 f + b_1) + b_2) \quad (1)$$

where Softplus ensures $\hat{y} > 0$, enforcing physically valid positive-valued predictions. The backbone is partially frozen during training: layers up to ResNet stage 2 are fixed (pretrained weights retained), while stages 3 and 4 are fine-tuned together with the regression heads. This strategy preserves low-level texture features learned from ImageNet while adapting high-level representations to crack imagery. Fig. 2 illustrates the full architecture.

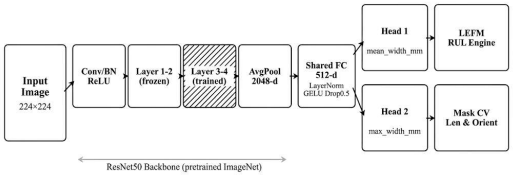

Fig. 2: DamCrackCNN architecture. Frozen layers shown in light green; trainable in dark green.

3.4 Training Protocol

The AdamW optimiser was used with differential learning rates: 5×10^{-6} for backbone parameters and 5×10^{-5} for the shared layer and regression heads, with L2 weight decay of 10^{-3}. A CosineAnnealingWarmRestarts scheduler with T_0 = 20 epochs provided periodic learning rate resets to

escape local minima — particularly beneficial for small datasets. The multi-task loss function:

$$L = \lambda_1 MSE(\bar{\hat{w}}_{tr}^{ue}, \bar{\hat{w}}_{tr}^{ue}) + \lambda_2 MSE(w_{tr}^{ue}, w_{tr}^{ue}) \quad (2)$$

was minimised with $\lambda_1 = \lambda_2 = 1.0$. Online augmentation (random crop, horizontal/vertical flip, colour jitter, random erasing after ToTensor) was applied during training. Mixed-precision (FP16) training on GPU reduced memory usage and accelerated training. A patience of 20 epochs was used for early stopping based on validation loss.

3.5 Crack Geometry from Masks

Crack length is estimated via skeletonisation of the binary segmentation mask. The distance transform:

$$D(x,y) = \min_{\partial M} \sqrt{(x-i)^2 + (y-j)^2} \quad (3)$$

is thresholded at 50% of its maximum to produce an approximate medial axis (skeleton). Crack length in pixels equals the skeleton pixel count. Crack orientation θ is derived from the second-order central moments $\mu^{p\varphi}$ of the crack pixel set via PCA:

$$\theta = \tfrac{1}{2} \tan^{-1}[2\mu_{11} / (\mu_{20} - \mu_{02})] \quad (4)$$

These classical CV estimates complement the CNN-predicted width values and feed directly into the fracture mechanics model.

3.6 Fracture Mechanics and RUL Estimation

Fig. 3 shows the LEFM computation chain. Hydrostatic water pressure at depth y below the reservoir surface is computed as :

$$p(y) = \rho g y \quad (5)$$

The nominal stress and normal stress on the crack plane (inclined at angle θ to the horizontal) are:

$$\sigma = p(y)/t, \qquad \sigma_n = \sigma \cos^2(\theta) \quad (6)$$

The Mode-I stress intensity factor is given as, assuming a surface crack with geometry factor Y:

$$K_I = Y \sigma_n \sqrt{(\pi a)} \quad (7)$$

Fracture occurs when $K_I \geq K_{Ic}$. The critical crack depth is:

$$a_crit = (K_{Ic} / Y\sigma_n)^2 / \pi \quad (8)$$

Crack growth under cyclic loading follows Paris' Law:

$$da/dt = C(K_I)^m \quad (9)$$

Numerical forward integration with daily time step $\Delta t = 86400$ s:

$$a_{t+\Delta t} = a_t + C(K_I)^m \cdot \Delta t \quad (10)$$

is continued until $K_I(a) \geq K_{Ic}$ or $a \geq a_crit$. RUL equals total elapsed simulation time converted to years. Table 3 lists all model parameters used.

Table 3: LEFM and Paris' Law model parameters.

Param	Value	Param	Value
H	100 m	K_{Ic}	1.0 MPa√m
t	12 m	Y	1.1
ρ	1000 kg/m³	C	10^{-12}
g	9.81 m/s²	m	2.5

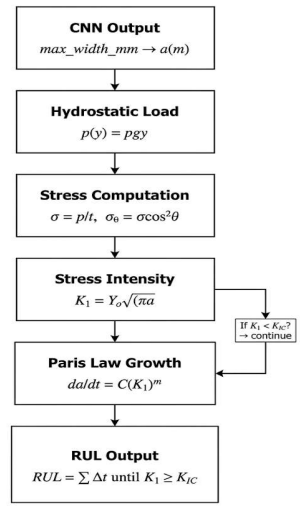

Fig. 3: LEFM and Paris' Law computation chain for RUL estimation.

4. Results and Discussion

4.1 Training Convergence

The model trained for all 80 epochs, and early stopping did not trigger. It reached the best validation loss of 1.355 at epoch 79. Fig. 4 shows both loss curves steadily decreasing without divergence. This confirms that the augmentation strategy successfully reduced overfitting, fitting model more accurately even with the small 45-image original dataset. The mean width head, with an MAE of 0.58 normalized at convergence, converged faster and more stably than the maximum width head, which had a very small MAE of 0.63.

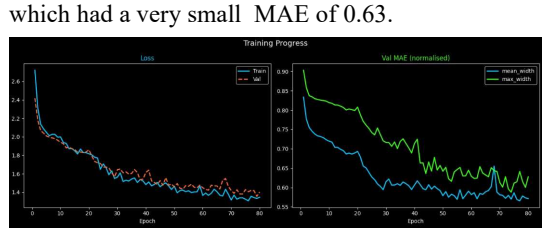

Fig. 4: Training/validation loss (left) and per-head MAE (right) over 80 epochs.

4.2 Width Prediction Accuracy

Table 1 reports test set accuracy evaluated on 45 images (9 original + augmented variants per test crack). The model achieves an MAE of 0.48 mm for mean width — clinically acceptable for cracks spanning the dataset range of 0.15 to 8.5 mm. The maximum width MAE of 1.27 mm is higher, reflecting the inherently greater variability of peak local width values across the five ROI positions. Fig. 5 presents predicted vs. true scatter plots for both outputs.

Table 1: CNN prediction accuracy on test set (45 images).

Target	MAE (mm)	RMSE (mm)
Mean Width	0.4805	0.7176
Max Width	1.2656	1.6321

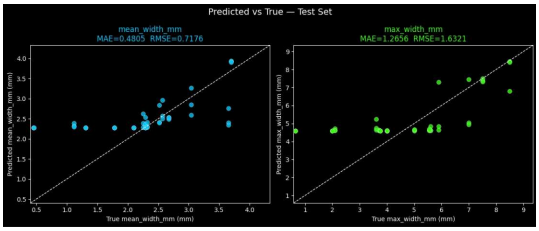

Fig. 5: Predicted vs. true crack widths on test set. Dashed line = perfect prediction.

Mean width predictions cluster near the perfect-prediction diagonal across the full range (Fig. 5, left), demonstrating that the model has learned a generalized mapping rather than simply memorizing training means. Maximum width predictions show wider scatter at values above 6 mm (Fig. 5, right), attributable to the small number of training examples with such extreme values. Both heads outperform a naive mean-prediction baseline (which would achieve MAE equal to the standard deviation of each target).

4.3 RUL Estimation

For the representative crack (mean width 2.29 mm, max width 4.0 mm, θ = 10°), the calculated values for the conservative water depth y = 40 m were:

$$\sigma_n = 0.0317 \text{ MPa}, \quad K_I = 0.00403 \text{ MPa}\sqrt{m} \quad (11)$$

This gives $K_I/K_{Ic} = 0.004$, thus the crack was classified as LOW risk. All 15 specimens were classified as LOW risk, with RUL > 50,000 years for the modelled hydrostatic loading. This makes physical sense, because the surface crack widths of 0.15–8.5 mm represent an early-stage surface deterioration, far below the critical crack depth ($a_{crit} \approx 261$ m) for fracture instability in the 12 m thick concrete dam wall. The approach is intended for use in situations where the crack depths are larger, the hydrostatic pressure on the dam wall is smaller, and the K_I/K_{Ic} ratios approach the critical values.

The low absolute values for K_I for this dataset notwithstanding, the relative values for the K_I/K_{Ic} ratio serve as a sensitive relative health index for the cracks, facilitating their ranking for inspection priority. The maximum width, used as the proxy for the crack penetration depth, was the dominant factor for the health index for the dataset, thus the importance of accurate prediction for the max width was emphasized.

5 Conclusion

This research successfully set up an automated system for measuring cracks in concrete dams and predicting their remaining useful life (RUL). It achieved a mean absolute error of 0.48 mm for average width and 1.27 mm for maximum width on test data. The modular design combines deep learning regression with physics-based fracture mechanics to deliver a reliable structural health index from standard field imagery. Future work will expand this framework firstly by including seismic, thermal, and uplift loading components relevant to dam environments and then combining the CNN outputs with detailed ANSYS simulations to replace simplified analytical LEFM engines. By using the Extended Finite Element Method (XFEM) and SMART technology in ANSYS, we aim to simulate complex, non-planar crack growth under both seismic and thermal loads. This move towards a real-time Digital Twin environment, supported by automated meshing and Python-based simulation control, will enable dynamic RUL updates and more accurate structural assessments as new inspection data comes in.

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